

APO-SERTRALINE
Sertraline Hydrochloride Capsules
Equivalent to 50mg Sertraline
Antidepressant

Pharmacology

The mechanism of action of sertraline is presumed to be linked to its ability to inhibit the neuronal reuptake of serotonin. It has only very weak effects on norepinephrine and dopamine neuronal reuptake. At clinical dose, sertraline blocks the uptake of serotonin into human platelets.

Like most clinically effective antidepressants, sertraline downregulates brain norepinephrine and serotonin receptors in animals. In receptor binding studies, sertraline has no significant affinity for adrenergic (alpha1, alpha2 & beta), cholinergic, GABA, dopaminergic, histaminergic, serotonergic (5-HT1A, 5-HT1B, 5-HT2) or benzodiazepine binding sites.

Pharmacokinetics

Following multiple oral once-daily doses of 200 mg, the mean peak plasma concentration (C_{max}) of sertraline is 0.19 µg/ml occurring between 6 to 8 hours post-dose. Following single or multiple oral once-daily doses of 50 to 400 mg/day the average terminal elimination half-life is approximately 26 hours. Linear dose proportionality has been demonstrated over the clinical dose range of 50 to 200 mg/day.

Food appears to increase the bioavailability by about 40%; it is recommended that APO-SERTRALINE (sertraline hydrochloride) be administered with meals.

Sertraline is extensively metabolized to N-desmethylsertraline, which shows negligible pharmacological activity. Both sertraline and N-desmethylsertraline undergo oxidative deamination and subsequent reduction, hydroxylation and glucuronide conjugation. Biliary excretion of metabolites is significant.

Approximately 98% of sertraline is plasma protein bound. The interactions between sertraline and other highly protein bound drugs have not been fully evaluated (see PRECAUTIONS).

The pharmacokinetics of sertraline itself appears to be similar in young and elderly subjects. Plasma levels of N-desmethylsertraline show a 3-fold elevation in the elderly following multiple dosing, however, the clinical significance of this observation is not known.

Indication

APO-SERTRALINE (sertraline hydrochloride) is indicated for the symptomatic relief of depressive illness.

Panic Disorder: APO-SERTRALINE is indicated for the symptomatic relief of panic disorder, with or without agoraphobia.

Obsessive-Compulsive Disorder: APO-SERTRALINE is indicated for the symptomatic relief of obsessive-compulsive disorder (OCD). The obsessions or compulsions must be experienced as intrusive, markedly distressing, time-consuming, or significantly interfering with the person's social or occupational functioning.

Recommended Dose

APO-SERTRALINE (sertraline hydrochloride) should be administered with food once daily preferably with the evening meal, or, if administration in the morning is desired, with breakfast.

Depression and Obsessive-Compulsive Disorder: As no clear dose-response relationship has been demonstrated over a range of 50-200 mg/day, a dose of 50 mg/day is recommended as the initial dose.

Panic disorder: APO-SERTRALINE treatment should be initiated with a dose of 25 mg once daily. After one week, the dose should be increased to 50 mg once daily depending on tolerability and clinical response. No clear dose-response relationship has been demonstrated over a range of 50-200 mg/day. **Titration:** In depression, OCD and panic disorder, a gradual increase in dosage may be considered if no clinical improvement is observed. Based on pharmacokinetic parameters, steady state sertraline plasma levels are achieved after approximately 1 week of once daily dosing; according, dose changes, if necessary, should be made at intervals of at least one week. Doses should not exceed a maximum of 200 mg/day.

The full therapeutic response may be delayed until 4 weeks of treatment or longer. Increasing the dosage rapidly does not normally shorten this latent period and may increase the incidence of side effects.

Maintenance: During long-term therapy for any indication, the dosage should be maintained at the lowest effective dose and patients should be periodically reassessed to determine the need for continued treatment.

Switching Patients To or From a Monoamine Oxidase Inhibitor: At least 14 days should elapse between discontinuation of an MAOI and initiation of therapy with APO-SERTRALINE. In addition, at least 14 days should be allowed after stopping APO-SERTRALINE before starting an MAOI (see CONTRAINDICATIONS).

Mode of administration: Oral administration

Contraindication

APO-SERTRALINE (sertraline hydrochloride) is contraindicated in patients with known hypersensitivity to the drug.

Monoamine Oxidase Inhibitors: Cases of serious, sometimes fatal, reactions have been reported in patients receiving sertraline in combination with a monoamine oxidase inhibitor (MAOI), including the

selective MAOI, selegiline and the reversible MAOI (reversible inhibitor of monoamine oxidase – RIMA), moclobemide. Some cases presented with features resembling the serotonin syndrome. Similar cases have been reported with other antidepressants during combined treatment with an MAOI and in patients who have recently discontinued an antidepressant and have been started on an MAOI. Symptoms of a drug interaction between an SSRI and an MAOI include: hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, mental status changes that include confusion, irritability, and extreme agitation progressing to delirium and coma. Therefore, sertraline should not be used in combination with an MAOI or within 14 days of discontinuing treatment with an MAOI. Similarly, at least 14 days should elapse after discontinuing sertraline treatment before starting an MAOI. Concomitant use in patients taking pimozide is contra-indicated (see section Drug Interaction). Sertraline should not be used in children and adolescents under the age of 18 years with Major Depressive Disorder.

Warnings

Monoamine Oxidase Inhibitors: see (CONTRAINDICATIONS).

Suicidality in Children and Adolescents:

- Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
- Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
- Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.
- Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.

The possibility of a suicide attempt is inherent in depression and may persist until significant remission occurs. Therefore, high risk patients should be closely supervised throughout therapy and consideration should be given to the possible need for hospitalization. In order to minimize the opportunity for overdose, prescriptions for APO-SERTRALINE should be written for the smallest quantity of drug consistent with good patient management.

Because of the well-established co-morbidity between both obsessive-compulsive disorder and depression and panic disorder and depression, the same precautions should be observed when treating patients with obsessive-compulsive disorder and panic disorder.

Seizures: Seizures are a potential risk with antidepressant or antiobsessional drugs. The drug should be discontinued in any patient who develops seizures. Sertraline should be avoided in patients with unstable epilepsy and patients with controlled epilepsy should be carefully monitored. Sertraline should be discontinued if there is an increase in seizure frequency.

Switching from SSRIs, Antidepressant and Antiobsessional: Care and prudent medical judgement should be exercised when switching, particularly from long-acting agents. The duration of washout period should intervene before switching from one selective serotonin reuptake inhibitor to another has not been established.

Sexual dysfunction: Selective serotonin reuptake inhibitors (SSRIs)/serotonin norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction. There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SSRIs/SNRIs.

Precautions

Activation of Mania/Hypomania: Activation of mania/hypomania has been reported in a small proportion of patients with Major Affective Disorder treated with other marketed antidepressants. Occupational Hazards: Any psychoactive drug may impair judgment, thinking, or motor skills, patients should be advised to avoid driving a car or operating hazardous machinery until they are reasonably certain that the drug treatment does not affect them adversely.

Hepatic Dysfunction: Serotonin is extensively metabolized by the liver. The use of sertraline in patients with hepatic disease must be approached with caution. If sertraline is administered to patients with hepatic impairment, a lower or less frequent dose should be considered.

Renal Dysfunction: Sertraline is extensively metabolized and excretion of unchanged drug in the urine is a minor route of elimination. It should be used with caution in renal impairment patients.

Use in Children: The safety and effectiveness of sertraline in children below the age of 18 have not been established.

Use in Elderly: The pattern of adverse reactions in the elderly was comparable to that in younger patients.

Drug Interactions

Monoamine Oxidase Inhibitors: see (CONTRAINDICATIONS).

CNS Active Drugs: Sertraline (200 mg daily) did not potentiate the effects of carbamazepine, haloperidol or phenytoin on cognitive and psychomotor performance in healthy subjects. Caution is advised if the concomitant administration of APO-SERTRALINE and such drugs is required. Serotonergic Drugs: Co-administration with

tryptophan may lead to a high incidence of serotonin-associated side effects. Serotonergic drugs, such as tramadol, sumatriptan or fenfluramine, should not be used concomitantly with sertraline, due to a possible 5-HT associated effects.

St. John's Wort: In common with other SSRI's, pharmacodynamic interactions between sertraline and the herbal remedy St. John's Wort may occur and may result in an increase in undesirable effects.

Lithium: As with other SSRIs, caution is recommended when coadministering sertraline with medications, such as lithium, which may act via serotonergic mechanisms.

Pimozide: While the mechanism of this interaction is unknown, due to the narrow therapeutic index of pimozide, concomitant of pimozide and sertraline is contra-indicated.

Alcohol: Although sertraline did not potentiate the cognitive and psychomotor effects of alcohol in experiments with normal subjects, the concomitant use of APO-SERTRALINE and alcohol in depressed, panic disorder or OCD patients has not been studied and is not recommended.

Cimetidine: The mean sertraline C_{max} and AUC were significantly higher in the cimetidine-treated group, as were the mean desmethylsertraline T_{max} and AUC. These data suggest that concomitant administration of cimetidine may inhibit the metabolism of sertraline and its metabolite, desmethylsertraline, and may result in a decrease in the clearance and first pass metabolism of sertraline, with a possible increase in drug-related side effects.

Warfarin: Prothrombin time should be carefully monitored when sertraline therapy is initiated or stopped in patients receiving warfarin.

Others: Sertraline did not potentiate the effects of carbamazepine, haloperidol or phenytoin on cognitive and psychomotor performance in healthy subjects.

Use in Pregnancy and Lactation: Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation.

The safety of sertraline during pregnancy and lactation has not been established and therefore, it should not be used in women of childbearing potential or nursing mothers, unless, in the opinion of the physician, the potential benefits to the patient outweigh the possible hazards to the fetus. There have been isolated reports of reactions such as tremors, jitteriness, restlessness, hypertonia,

hyperreflexia and difficulty breathing in neonates whose mothers had been treated with sertraline during pregnancy and in an infant whose mother discontinued sertraline treatment during breast feeding. However, causal relationship between sertraline treatment and the emergence of these events has not been established.

Observational data indicate an increased risk (less than 2-fold) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.

Adverse Effects

The most commonly observed adverse events associated with the use of sertraline were: Nausea, diarrhea/loss stool, anorexia, dyspepsia, male sexual dysfunction (principally ejaculatory delay); insomnia and somnolence; tremor; increased sweating and dry mouth and dizziness. In paediatric OCD patients, side effects which occurred more frequently with sertraline than placebo were: headache, insomnia, agitation, anorexia, and tremor. Post-marketing reports include the following:

Cardiovascular: Blood pressure disturbances including postural hypotension, tachycardia.

Gastrointestinal disorders: Vomiting, abdominal pain, microscopic colitis/ colitis microscopic*

*ADR identified post-marketing

Eye disorders: Abnormal vision.

Nervous system: Amnesia, headache, drowsiness, movement disorders, paraesthesia, hypoaesthesia, depressive symptoms, hallucinations, aggressive reaction, agitation, anxiety, psychosis, nervousness, panic reaction, fever, rigidity, confusion, diaphoresis, tachycardia, hypertension and diarrhea. Manic reaction has been reported.

Seizures: Sertraline should be discontinued in any patient who develops seizures.

Musculoskeletal: Arthralgia, myalgia.

Hepatic/pancreatic: Rarely, pancreatitis and serious liver events. Asymptomatic elevations in serum transaminases (SGOT and SGPT) have been reported, with an increased risk with the 200mg daily dose. These abnormalities occurred within the first 1 to 9 weeks of drug treatment and promptly diminished upon drug discontinuation.

Renal disorders: Urinary retention.

Reproductive: Hyperprolactinemia, galactorrhea, menstrual irregularities, anorgasmia.

Skin and allergic reactions: Rashes, angioedema, pruritus, ecchymoses and anaphylactoid reactions.

Metabolic: Rarely, hyponatremia.

Haematologic: Altered platelet function and/or abnormal clinical laboratory results have been reported rarely. Thrombocytopenia, abnormal bleeding or purpura in several patients taking sertraline has been reported but it is unclear whether sertraline had a causative role.

General: Malaise.

Other: Withdrawal symptoms include dizziness, paraesthesia, headache, anxiety and nausea have

been reported. Abrupt discontinuation of sertraline should be avoided.

Symptoms and Treatment of Overdose

Overdoses of sertraline alone up to 6g have been reported. Symptoms of overdose with sertraline alone included somnolence, nausea, vomiting, tachycardia, ECG changes, anxiety and dilated pupils. Treatment was primarily supportive and included monitoring and use of activated charcoal, gastric lavage or cathartics and hydration.

Availability of Dosage Forms

APO-SERTRALINE 50 mg: White opaque body, yellow opaque cap, hard gelatin capsule, imprinted APO 50. Homogenous white to off-white powder fill. Available in HDPE bottles of 100.

Storage Conditions

Apo-Sertraline (sertraline hydrochloride) should be stored below 30°C.

Manufacturer

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